Simplifying Radical Expressions

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Learning Objectives

- Identify perfect squares and use radical notation correctly
- Simplify square roots of non-perfect (irregular) numbers using factor trees
- Simplify radical expressions containing variables
- Apply factoring skills to fully simplify radical expressions

Simplify each radical expression completely, showing all work and leaving answers in simplest radical form.

1. Evaluate the perfect square root.

$\sqrt{81}$

Answer: ________________________________

2. Evaluate the perfect square root.

$\sqrt{144}$

Answer: ________________________________

3. Simplify the irregular square root using a factor tree.

$\sqrt{72}$

Answer: ________________________________

4. Simplify the irregular square root using a factor tree.

$\sqrt{50}$

Answer: ________________________________

5. Simplify the irregular square root.

$\sqrt{200}$

Answer: ________________________________

6. Simplify the radical expression with a variable.

$\sqrt{x^6}$

Answer: ________________________________

7. Simplify the radical expression with variables.

$\sqrt{49x^4y^2}$

Answer: ____________________________________

8. Simplify the radical expression with variables.

$\sqrt{18x^3}$

Answer: ____________________________________

9. Simplify the radical expression with variables.

$\sqrt{75a^5b^2}$

Answer: ____________________________________

10. Simplify the radical expression completely.

$\sqrt{128x^7y^4}$

Answer: ____________________________________

Covers warm-up review, introduction to radicals and radical notation, perfect squares, irregular square roots, simplifying radical expressions, radicals with variables, factor-tree method, and applying factoring skills to fully simplify radical expressions.

Solutions

1. Evaluate the perfect square root.

$\sqrt{81}$

→ *Recognize 81 as a perfect square.*
→ *Since 9 times 9 equals 81, the square root of 81 is 9.*

Answer: 9

2. Evaluate the perfect square root.

$\sqrt{144}$

→ *Recognize 144 as a perfect square.*
→ *Since 12 times 12 equals 144, the square root of 144 is 12.*

Answer: 12

3. Simplify the irregular square root using a factor tree.

$\sqrt{72}$

→ *Factor 72 into prime factors: 2, 2, 2, 3, 3.*
→ *Pair the like factors: a pair of 2s and a pair of 3s, with one 2 left over.*
→ *Each pair leaves the radical as a single factor, giving 2 times 3 equals 6 outside the radical.*
→ *The leftover 2 stays inside the radical, giving six root two.*

Answer: $6\sqrt{2}$

4. Simplify the irregular square root using a factor tree.

$\sqrt{50}$

→ *Factor 50 as 25 times 2.*
→ *Since 25 is a perfect square, take its square root, which is 5.*
→ *Bring 5 outside the radical and keep 2 inside, giving five root two.*

Answer: $5\sqrt{2}$

5. Simplify the irregular square root.

$\sqrt{200}$

→ *Factor 200 as 100 times 2.*
→ *Since 100 is a perfect square, take its square root, which is 10.*
→ *Place 10 outside the radical and keep 2 inside, giving ten root two.*

Answer: $10\sqrt{2}$

6. Simplify the radical expression with a variable.

$\sqrt{x^6}$

→ *Apply the rule that the square root of x to an even power equals x raised to half that power.*
→ *Divide the exponent 6 by 2 to get 3.*
→ *The simplified expression is x cubed.*

Answer: x^3

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7. Simplify the radical expression with variables.

$\sqrt{49x^4y^2}$

→ *Take the square root of 49 to get 7.*
→ *Take the square root of x to the fourth, which is x squared.*
→ *Take the square root of y squared, which is y.*
→ *Multiply the results to get seven x squared y.*

Answer: $7x^2y$

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8. Simplify the radical expression with variables.

$\sqrt{18x^3}$

→ *Factor 18 as 9 times 2 and rewrite x cubed as x squared times x.*
→ *Take the square root of 9, which is 3, and the square root of x squared, which is x.*
→ *Place 3 and x outside the radical, leaving 2 and x inside.*
→ *The simplified expression is three x times the square root of two x.*

Answer: $3x\sqrt{2x}$

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9. Simplify the radical expression with variables.

$\sqrt{75a^5b^2}$

→ *Factor 75 as 25 times 3.*
→ *Rewrite a to the fifth as a to the fourth times a.*
→ *Take the square root of 25 to get 5, the square root of a to the fourth to get a squared, and the square root of b squared to get b.*
→ *Leave 3 and the single a inside the radical, giving five a squared b times the square root of three a.*

Answer: $5a^2b\sqrt{3a}$

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10. Simplify the radical expression completely.

$\sqrt{128x^7y^4}$

→ *Factor 128 as 64 times 2.*
→ *Rewrite x to the seventh as x to the sixth times x.*
→ *Take the square root of 64 to get 8, the square root of x to the sixth to get x cubed, and the square root of y to the fourth to get y squared.*
→ *Leave 2 and the single x inside the radical, giving eight x cubed y squared times the square root of two x.*

Answer: $8x^3y^2\sqrt{2x}$